HS-OBS	Functional Requirements Specification
DEPARTMENT	DOCUMENT TYPE

VZI R1

Operating Modes Functional Requirements Specification Internal doors

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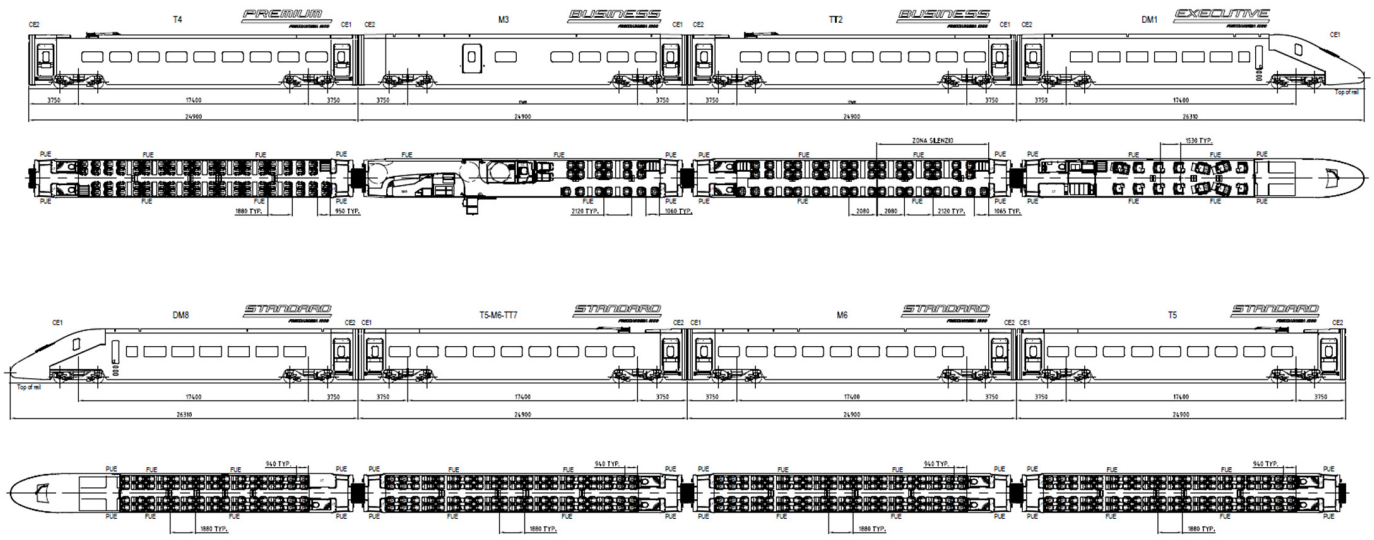
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1 Introduction

This document is a function specification for the control of the Interior doors system for V300ZEFIRO for customer Trenitalia.

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

CD	Circuit Diagram
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDU	Interface Display Unit
TCMS	Train Control and Management System
FDS	Fire Detection System
DCU	Door Control Unit
SW	Software

2 System Overview

The interior doors system includes automatic saloon doors and gangway doors.

Automatic saloon doors

It is a single leaf sliding door electrically driven. There are two kinds of saloon door (STD saloon door and HK saloon door) differing only for the opening free width (700mm Vs 900mm)

For each saloon door a dedicated Door Control Unit (DCU) is foreseen. The DCU manages all the door functionalities, particularly:

- opening command (by presence sensors and by push buttons on both side of the door)
- closing command (by internal timer)
- passenger protection system (jamming protection)
- diagnostic

Among the STD saloon doors, the 2 doors in DM1 coach confining the galley area are called Galley doors and differ from the STD saloon doors for the fact that presence sensors are installed only on the passenger saloon and vestibule side to prevent that the presence of galley area staff result in a permanent door open command. Moreover, a dedicated function allows the synchronization in the opening of the 2 doors to improve passenger comfort.

Quantities and distribution of the internal doors on the train are shown in the *Figure 1 – Internal doors identification/layout.*

Each saloon door is connected to the TCMS by means of two hardwired lines (for diagnostic purpose only).

Gangway door

The gangway door of the M3 coach on the bistro side (indicated as GWDoor_R_M3) it is a normally closed double leaf sliding door electrically driven provided with force closing device (spring). The dedicated Door Control Unit (DCU) manages all the door functionalities (as saloon doors), particularly:

- opening command (by push buttons on both side of the door)
- closing command (by internal timer)
- passenger protection system (jamming protection)
- diagnostic
- opening inhibition function (fire protection)

In case of fire the door will close (if open) and remain closed by cutting the power supply to the control unit. When closed, the door is operable by hand.

This door can be locked in its fully open position in case of any failure or for maintenance. If locked in open position, the door has the same behaviour of the standard gangway doors.

Silence doors

It is a normally close single leaf sliding door electrically driven, installed in the middle of TT2 cars for border the silence area as shown in the Figure 1 – Internal doors identification/layout.

For each saloon door a dedicated Door Control Unit (DCU) is foreseen. The DCU manages all the door functionalities, particularly:

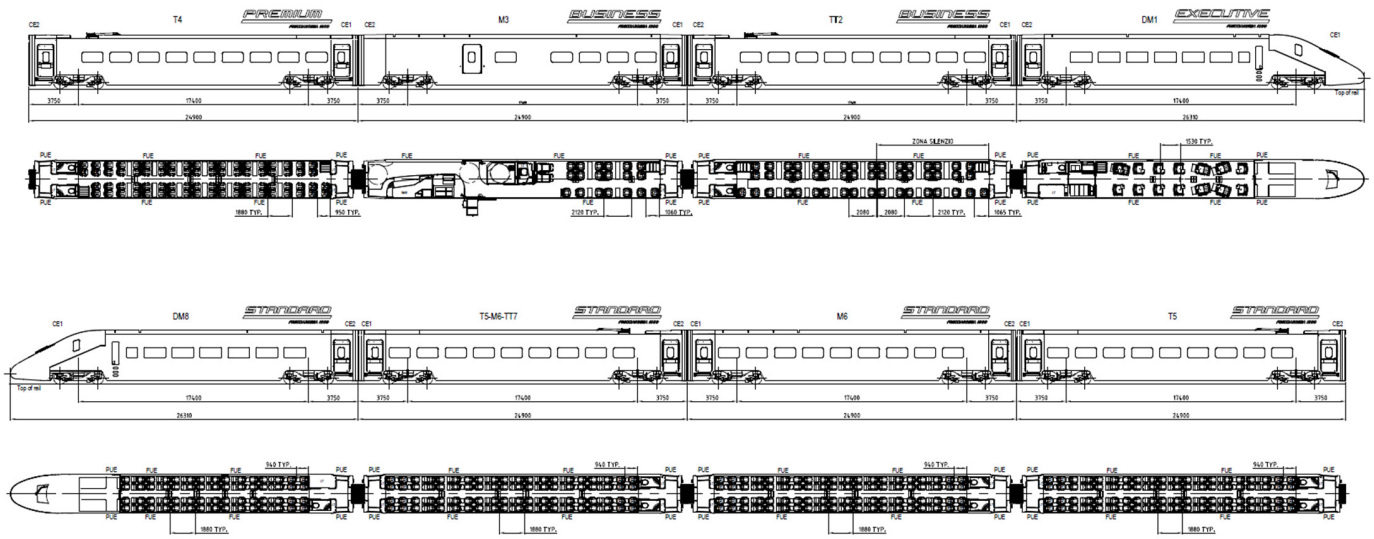
- opening command (by presence sensors and by push buttons on both side of the door)
- closing command (by internal timer)
- passenger protection system (jamming protection)
- diagnostic

Silence door is connected to the TCMS by means of two hardwired lines (for diagnostic purpose only).

This door can be locked in its fully open position in case of any failure or for maintenance.

2.1 System Configuration

The internal doors are identified according to the following rule:



3 Functional requirements

3.1 Gangway Doors

3.1.1 Gangway Door - Fire mode

3.1.1.1 Normally open gangway doors

Following sub sections are applicable to all the standard gangway doors and to the GWDoor_R_M3 if it has been locked in open position.

3.1.1.1.1 Function Design

The gangway door closing command is realized by means of hardwired line defined in Fire Detection System (FDS) SFRS. The logic, in terms of when door hast to be closed in case of fire, is handled by the fire detection system. Gangway door will close as soon as the relevant closing command is energized. In parallel to the FDS command a thermal fuse is able to close the door if the temperature near the gangway door is greater than a predefined threshold

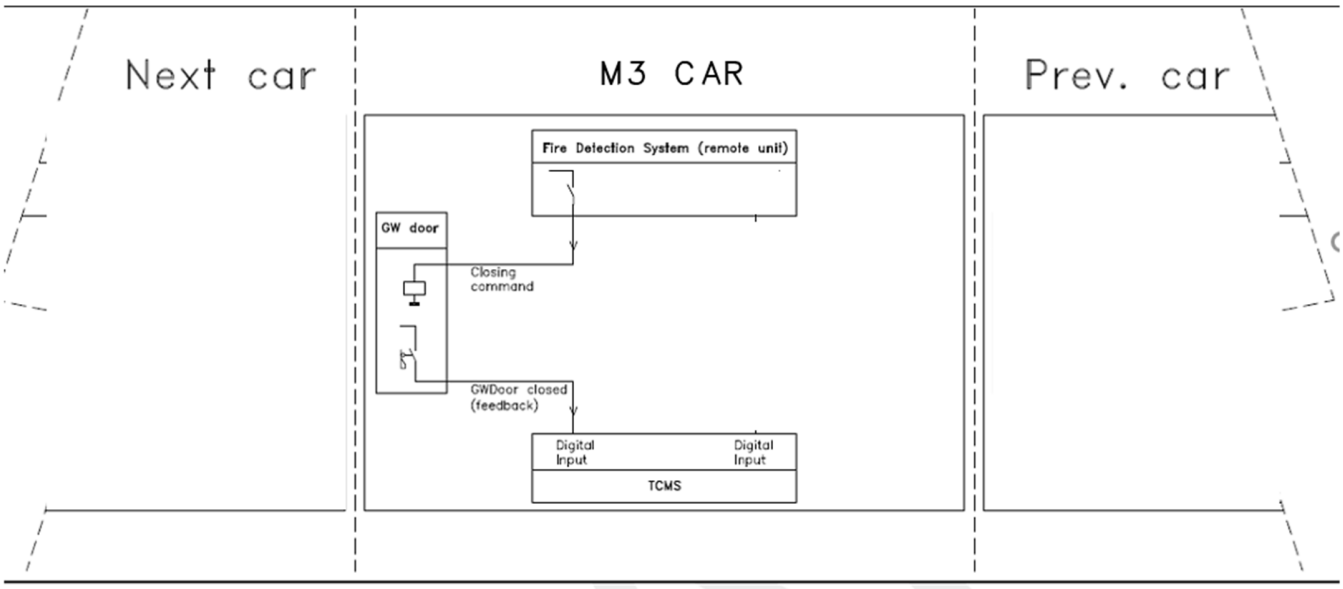


Figure 2 - Gangway door closing command block diagram (coach level)

3.1.1.1.2 Normal condition

In case of fire, when the gangway door receives the relevant command from the FDS, the locking device (electrical motor) is released and the closing device (spring) closes the door leafs.

If the temperature near to the gangway door mechanism is above a pre-set value, a thermal fuse melts and releases the locking device starting the closing sequence, independently by the FDS status.

3.1.1.1.3 Failure cases

a) Loss of function of the locking device (electrical motor). Loss of a redundancy (the thermal fuse is still able to close the door).
b) Thermal fuse doesn't melt but over temperature is present. Loss of a redundancy (the FDS is still able to close the door).
c) Thermal fuse melts but no over temperature is present. Undue closing of the door (it can be opened manually)
d) Loss of function of the closing device (spring). The gangway doesn't close.

3.1.1.1.4 Failure detection

a) Failure condition detected by the TCMS comparing the FDS command with the current position of the gangway door
b) Failure condition detected by the TCMS comparing the FDS command with the current position of the gangway door
c) Failure condition detected by the TCMS comparing the FDS command with the current position of the gangway door
d) Failure condition detected by the TCMS comparing the FDS command with the current position of the gangway door

3.1.1.1.5 Failure handling

a) Driver has to act according to operational manual
b) Driver has to act according to operational manual
c) Melted thermal fuse has to be changed during maintenance.
d) Driver has to act according to operational manual

3.1.1.2 Normally closed gangway door

Following sub sections are applicable to the GWDoor_R_M3 in its normal functioning, i.e. when the door is not locked.

3.1.1.2.1 Function Design

The closing command is realized by means of an hardwired signal defined in the Fire Detection System (FDS) SFRS.
The need of closing (if open) or keeping this door closed (if already closed) in case of fire is handled by the fire detection system.

As soon as the relevant command from the fire detection system is energized, the power supply will be cut to the DCU by means of hard wired circuits in such a way that the DCU can not command any door movement. As a redundancy, the DCU SW will inhibit all the undue door opening command.

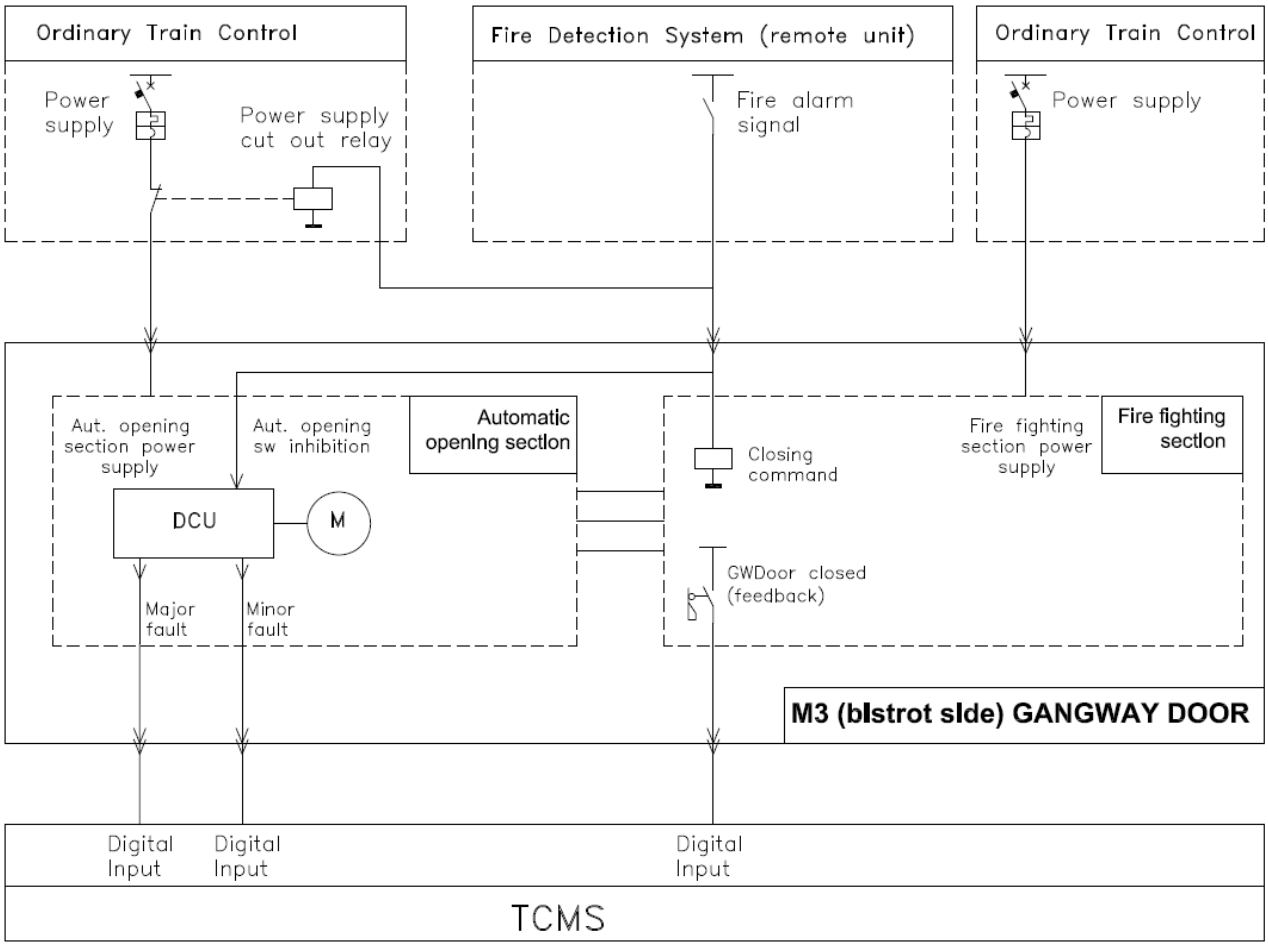


Figure 3 – M3 rear gangway door schematic

3.1.1.2.2 Normal condition

In normal condition the function is redundantly realized by means of HW components and DCU SW.

3.1.1.2.3 Failure cases

a) Loss of function of the HW power supply cut off. Loss of a redundancy (the DCU SW still inhibits the door opening).

b) Loss of function of the DCU SW inhibition function. Loss of a redundancy (the HW power supply cut off prevents the door from opening).

3.1.1.2.4 Failure detection

a) Failure condition detected by the TCMS comparing the FDS command with the major fault signal of the gangway door.
b) Not detected.

3.1.1.2.5 Failure handling

a) Driver has to act according to operational manual.

3.1.1.3 Requirements

ID	
Description	Gangway door of the car where a fire has been detected shall automatically close.
Safety level	SIL 1

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.04.Zefiro-Europe.CONCEPT.61	The gangway door shall receive the closing command by means of a hardwired line.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_08.04.Zefiro-Europe.CONCEPT.97	The gangway door shall close when the closing command hardwired line is energized	Derived Requirement	4S_08.06.02.-.Gangway Doors		SIL 1
2F_08.04.Zefiro-Europe.CONCEPT.130	Power supply shall be cut to the GWDoor_R_M3 DCU as soon as the command from the fire detection system is energized.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_08.04.Zefiro-Europe.CONCEPT.131	DCU SW shall inhibit any door movement as soon as the command from the fire detection system is energized.	Derived Requirement	4S_08.06.02.-.Gangway Doors		SIL 0

3.2 Galley Doors

3.2.1 Function design

The DCUs of the 2 saloon doors in DM1 coach are connected one to the other in order to realize a synchronization function.

3.2.2 Normal condition

Once one of the 2 door is commanded to open, by the presence sensors or the bush button, the DCU of such door will command the DCU of the other door to open.

3.2.3 Failure cases

a) Loss of the synchronization function. The door can still be open by the presence sensor or the push button.

3.2.4 Failure detection

a) Not detected.

3.2.5 Failure handling

NA

3.2.6 Requirements

ID	2F_08.04.Zefiro-Europe.TRS.84
Description	The 2 galley doors shall be synchronized and operated in parallel by the activation of one of the presence sensors or by one of the pushbuttons.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.04.Zefiro-Europe.CONCEPT.145	A suitable hardwired connection shall be foreseen in order to transfer the synchronization signals.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_08.04.Zefiro-Europe.CONCEPT.146	Once one of the 2 galley door is commanded to open, by the presence sensors or the bush button, the DCU of such door shall command the DCU of the other galley door to open	Derived Requirement	4S_08.06.-.Interior Doors		SIL 0

3.3 Silence Door

3.3.1 Function design

Interior door for silence area shall be provided with door panel on which shall be installed (on both side) door open push button. No photocell shall be installed.

3.3.2 Normal condition

In normal condition the function is redundantly realized by means of HW components and DCU SW.

3.3.3 Failure cases

NA

3.3.4 Failure detection

Not detected.

3.3.5 Failure handling

NA

3.3.6 Requirements

See following paragraphs.

3.4 Supervision

3.4.1 Diagnostic data

3.4.1.1 Function Design

Saloon, silence and gangway doors are connected to the TCMS for diagnostic purpose using hardwired lines (two lines for each saloon and silence door and one line for each gangway door).
Each saloon door and the GWDoor_R_M3 provide information about its actual status sending to the TCMS the following signals:

- Major fault (door out of order). This signal is normally at high level (+Batt level means door OK)
- Minor fault (the door is still able to work but maintenance is needed). This signal is normally at low level (+Batt level means maintenance needed)

Each gangway door provides information about its actual status sending to the TCMS the following signal:

- Closed position. This signal is normally at low level (+Batt means door closed)

This information is compared by the TCMS with the relevant closing command coming from the FDS in order to identify unexpected scenarios like:

- gangway door unexpected closed (without FDS command)
- gangway closing command failed (the FDS commands the closing of the gangway door due to a fire and the gangway door does not close)

3.4.1.2 Requirements

ID	2F_08.04.Zefiro-Europe.TRS.63
Description	Diagnostic data exchange between TCMS and internal doors shall be done by means of hardwired lines.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.04.Zefiro-Europe.CONCEPT.66	A major failure on a saloon door (SDoor_y_zzz_MjFault) shall be transmitted to TCMS	Derived Requirement	4S_08.06.-.Interior Doors		SIL 0
2F_08.04.Zefiro-Europe.CONCEPT.99	The major failure of each saloon door (SDoor_y_zzz_MjFault) shall be transmitted to TCMS by means of a dedicated wire.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_08.04.Zefiro-Europe.CONCEPT.100	TCMS shall store an event (Warning) in case of a major fault of a saloon door (SDoor_y_zzz_MjFault).	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_08.04.Zefiro-Europe.CONCEPT.104	A minor failure on a saloon door (SDoor_y_zzz_MnFault) shall be transmitted to TCMS	Derived Requirement	4S_08.06.-.Interior Doors		SIL 0
2F_08.04.Zefiro-Europe.CONCEPT.105	The minor failure of each saloon door (SDoor_y_zzz_MnFault) shall be transmitted to TCMS by means of a dedicated wire	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_08.04.Zefiro-Europe.CONCEPT.101	TCMS shall store an event (Maintenance) in case of a minor fault of a saloon door (SDoor_y_zzz_MnFault).	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_08.04.Zefiro-Europe.CONCEPT.147	A major failure on the GWDoor_R_M3 (GWDoor_R_M3 _MjFault) shall be transmitted to TCMS	Derived Requirement	4S_08.06.02.-.Gangway Doors		SIL 0
2F_08.04.Zefiro-Europe.CONCEPT.148	The major failure of the GWDoor_R_M3 (GWDoor_R_M3 _MjFault) shall be transmitted to TCMS by means of a dedicated wire.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_08.04.Zefiro-Europe.CONCEPT.149	TCMS shall store an event (Warning) in case of a major fault of the GWDoor_R_M3 (GWDoor_R_M3 _MjFault).	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_08.04.Zefiro-Europe.CONCEPT.150	A minor failure on the GWDoor_R_M3 (GWDoor_R_M3 _MnFault) shall be transmitted to TCMS	Derived Requirement	4S_08.06.02.-.Gangway Doors		SIL 0
2F_08.04.Zefiro-Europe.CONCEPT.151	The minor failure on the GWDoor_R_M3 (GWDoor_R_M3 _MnFault) shall be transmitted to TCMS by means of a dedicated wire	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_08.04.Zefiro-Europe.CONCEPT.152	TCMS shall store an event (Maintenance) in case of a minor fault of the GWDoor_R_M3 (GWDoor_R_M3 _MnFault).	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_08.04.Zefiro-Europe.CONCEPT.169	A major failure on a silence door (SilenceDoor_y_zzz_MjFault) shall be transmitted to TCMS	New Requirement			SIL 0
2F_08.04.Zefiro-Europe.CONCEPT.170	The major failure of each silence door (SilenceDoor_y_zzz_MjFault) shall be transmitted to TCMS by means of a dedicated wire.	New Requirement			SIL 0
2F_08.04.Zefiro-Europe.CONCEPT.171	TCMS shall store an event (Warning) in case of a major fault of a silence door (SilenceDoor_y_zzz_MjFault).	New Requirement		EVENT	N/A
2F_08.04.Zefiro-Europe.CONCEPT.172	A minor failure on a silence door (SilenceDoor_y_zzz_MnFault) shall be transmitted to TCMS	New Requirement			SIL 0
2F_08.04.Zefiro-Europe.CONCEPT.173	The minor failure of each silence door (SilenceDoor_y_zzz_MnFault) shall be transmitted to TCMS by means of a dedicated wire	New Requirement			SIL 0
2F_08.04.Zefiro-Europe.CONCEPT.174	TCMS shall store an event (Maintenance) in case of a minor fault of a silence door (SilenceDoor_y_zzz_MnFault).	New Requirement		EVENT	N/A

Information about the gangway door status (closed position) is identified and sent to the TCMS at diagnostic purpose.
In addition, gangway door closing command (from the FDS) is sent to the TCMS in order to compare it with the actual status of the gangway door and identify unexpected scenarios

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.04.Zefiro-Europe.CONCEPT.108	The gangway door closed position (GDoor_R_DM1_Closed) shall be identified and sent to the TCMS.	Derived Requirement	4S_08.06.02.-.Gangway Doors		N/A
2F_08.04.Zefiro-Europe.CONCEPT.109	The gangway door closed position (GDoor_R_DM1_Closed) shall sent to the TCMS by means of a hardwired line	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_08.04.Zefiro-Europe.CONCEPT.110	TCMS shall store an event (Alarm) if the gangway is not closed 10 TBD seconds after the closing command from the FDS.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_08.04.Zefiro-Europe.CONCEPT.153	TCMS shall store an event (Maintenance) if the signal GWDoor_R_M3 _MjFault is at +Batt level while the command from the fire detection system is active for that door.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

4. Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_08.04.Zefiro-Europe.CONCEPT.28	GWDoor_y_ZZZ_ClsCmd Closing command from FDS		2F_08.04.Zefiro-Europe.CONCEPT.97

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_08.04.Zefiro-Europe.CONCEPT.29	SDoor_y_zzz_MjFault (Saloon door Major Fault)		2F_08.04.Zefiro-Europe.CONCEPT.66
2F_08.04.Zefiro-Europe.CONCEPT.112	Sdoor_y_zzz_MnFault (Saloon door Minor Fault)		2F_08.04.Zefiro-Europe.CONCEPT.104
2F_08.04.Zefiro-Europe.CONCEPT.111	GWdoor_y_zzz_Closed (Gangway door closed position)		2F_08.04.Zefiro-Europe.CONCEPT.108
2F_08.04.Zefiro-Europe.CONCEPT.154	GWDoor_R_M3 _MjFault (Gangway door Major Fault)		2F_08.04.Zefiro-Europe.CONCEPT.147
2F_08.04.Zefiro-Europe.CONCEPT.155	GWDoor_R_M3 _MnFault (Gangway door Minor Fault)		2F_08.04.Zefiro-Europe.CONCEPT.150
2F_08.04.Zefiro-Europe.CONCEPT.175	SilenceDoor_y_zzz_MjFault (Silence door Major Fault)		
2F_08.04.Zefiro-Europe.CONCEPT.176	Silencedoor_y_zzz_MnFault (Silence door Minor Fault)		

4.1.2 TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_08.04.Zefiro-Europe.CONCEPT.30	SDoor_y_zzz_MjFault (Saloon door Major Fault)		2F_08.04.Zefiro-Europe.CONCEPT.100
2F_08.04.Zefiro-Europe.CONCEPT.114	Sdoor_y_zzz_MnFault (Saloon door Minor Fault)		2F_08.04.Zefiro-Europe.CONCEPT.101
2F_08.04.Zefiro-Europe.CONCEPT.113	GWdoor_y_zzz_Closed (Gangway door closed position)		2F_08.04.Zefiro-Europe.CONCEPT.109
2F_08.04.Zefiro-Europe.CONCEPT.156	GWDoor_R_M3 _MjFault (Gangway door Major Fault)		2F_08.04.Zefiro-Europe.CONCEPT.149
2F_08.04.Zefiro-Europe.CONCEPT.157	GWDoor_R_M3 _MnFault (Gangway door Minor Fault)		2F_08.04.Zefiro-Europe.CONCEPT.152
2F_08.04.Zefiro-Europe.CONCEPT.177	SilenceDoor_y_zzz_MjFault (Silence door Major Fault)		
2F_08.04.Zefiro-Europe.CONCEPT.178	Silencedoor_y_zzz_MnFault (Silence door Minor Fault)		

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_08.04.Zefiro-Europe.CONCEPT.31	--		

4.2 Bus Interface

Supplier to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_08.04.Zefiro-Europe.CONCEPT.32	--		

TCMS SW to Supplier:

Nr	Functional name	Type	Linked to requirement
2F_08.04.Zefiro-Europe.CONCEPT.33	--		

5. References

[1] EC01P006798B TRD Interior Doors System

[2] EC01P006740B TRD Gangway Door System

[3] EC010030702 TRD Silence Door System